JAVOKHIR JAMBULOV

SOFTWARE ENGINEER | FULL-STACK, AI & SYSTEMS

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PROFILE

Software engineer with 2+ years of production experience across mobile, web, cloud infrastructure, and AI inference for Soundcorset (50M+ downloads; 400K+ daily active users). Built cross-platform integrations, asynchronous media systems, and on-device/server-side model inference; strong ownership from implementation to production reliability.

PROFESSIONAL EXPERIENCE

Software Engineer | OnSquare (Soundcorset) Aug 2024 – Present

Consumer music platform | Kotlin Android client, TypeScript web product, Scala/AWS backend

AI INFERENCE & MEDIA SYSTEMS

- **On-device AI:** Migrated production inference from PyTorch Mobile to ExecuTorch and extended multichannel audio preprocessing for AI Tutor, including audio format adaptation.
- **Server-side AI:** Built an end-to-end audio recording-evaluation pipeline integrating ONNX inference, S3 media storage, SQS processing jobs, and database persistence to return evaluation results to clients.

BACKEND, CLOUD & SYSTEM RELIABILITY

- **Distributed consistency:** Prevented deleted media records from resurfacing in asynchronous S3/SQS/DynamoDB workflows using conditional writes, DynamoDB Streams, existence checks, and ScalaTest race-condition tests.
- **Secure integration:** Implemented Google/Facebook authentication across the web client, Android WebView, and Scala APIs with token validation, server sessions, credentialed CORS, and role-based admin controls.
- **Production operations:** Added retryable uploads, per-user HTTP 429 rate limiting, browser-to-server error reporting, and GCP-backed moderation workflows with administrative controls.

WEB, CROSS-PLATFORM & CLIENT PERFORMANCE

- **Web platform:** Modernized legacy JavaScript/jQuery to strict TypeScript and Vite with modular features, ESLint, dependency cleanup, and circular-import checks.
- **Client integration:** Built pressure-sensitive sheet-music annotation with Canvas/OffscreenCanvas, a shared Web Worker, offline IndexedDB storage, and S3 sync across web and Android WebView.
- **Mobile performance:** Reduced slow cold-start incidence from ~10% to ~4.5% using Perfetto profiling; diagnosed crashes, ANRs, and lifecycle leaks, helping achieve a 99.98% crash-free user rate.

TECHNICAL SKILLS

Languages & client engineering: TypeScript/JavaScript, Kotlin, Scala, React, Next.js, Jetpack Compose, Android SDK, ViewModel, DI (Hilt), Coroutines/Flow

Backend & infrastructure: Akka HTTP, REST APIs, OAuth 2.0/JWT, AWS S3/SQS/DynamoDB, DynamoDB Streams, Terraform, Nginx, Caffeine

AI, web graphics & tooling: ExecuTorch, PyTorch Mobile, ONNX inference, Canvas/OffscreenCanvas, Web Workers, IndexedDB, Perfetto, Crashlytics, ScalaTest, Git

Robotics & foundational languages: ROS 2 with TurtleBot (academic); Python and C++ (basic)

INDEPENDENT PROJECTS

Published Android apps: Pomodoro: Focus Timer (Kotlin/Compose; 1K+ installs, ~100 DAU) and Take Notes (Kotlin, Room, MVVM).

EDUCATION & ACADEMIC ROBOTICS

Inha University | B.S. Integrated System Engineering, 2024 | GPA 4.28/4.5, rank 2/35, five academic awards

Academic robotics coursework: Worked with ROS 2 and TurtleBot mobile robots during university projects.

Languages: English (IELTS 7.5) | Korean (TOPIK Level 3)